Lecture Notes on QFT

October 20, 2025

Readings

- Peskin & Schroeder Sec. 2.1 and 2.2
- Weinberg vol 1 Chap. 1

1 Lecture 1: Classical Field Theories and Principle of Locality

The goals of this Quantum Field Theory class is

- To develop concepts, formalism and techniques for quantum dynamics of fields.
- Quantum Field Theory is also very important for other interactions in nature. Among the four fundamental interactions in nature, three of them are described by QFT completely.
- Even though QFT is initially developed for particle physics, but over the years, QFT has been found many applications in many other branches of physics, in condensed matter, statistical physics and so on. It's fair to say, nowadays, QFT has become a universal language for theoretical physics in all different fields.

2 Why we consider QFT?

2.1 principle of locality

The first important concept is called the **principle of locality**.

In Newtonian mechanics, you have action at a distance, for example, if you look at gravity and the gravity is exerted by the sun on the Earth, but they are very far away. And the same thing with the Coulomb interaction between the charged particles.

But then, in the 19th century, they came from this principle of locality, that is formulated by Farady around 1830. The principle of locality says that all points you don't have action at a distance. He said **all points in space participate in the physical process, and the effect propagates from points to a neighboring points**. So in this principle of locality, you don't have action at a distance, action at a distance is always conveyed from one point to another point through the propagating in the space.

The concepts of **fields** is **the mathematical device or vehicle that the principle locality is at work**. The main idea of the filed is that we associate each point in space a dynamical variable.

For example, if you have electric field is defined at each point x, we can introduce an electric field, which can also depend on time. Of course, normally we write it this way $\vec{E}_{\vec{x}}(t) \equiv \vec{E}(\vec{x}, t)$. The reason I write it in this way $\vec{E}_{\vec{x}}(t)$ to emphasize that in the definition of the electric field, the space and times actually play very different roles. The space plays the role of a label. So at each point $\vec{x}$, we have electric field, so x here is just a label, and t is used to describe the evolution, the change of the electric field. So $\vec{x}$ and t play very different roles.

Similarly, you can do this to magnetic field $\vec{B}_{\vec{x}}(t) \equiv \vec{B}(\vec{x}, t)$, you should always treat $\vec{x}$ as labels. And we know the evolution of electric and magnetic field is described by Maxwell's equations,

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{E} = 4\pi\rho \\ \frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} \\ \nabla \times \vec{B} = 4\pi\vec{J} + \frac{\partial \vec{E}}{\partial t} \end{cases} \tag{1}$$

which is called the differential form of the Maxwell's equations. This set of equations examplifies perfectly the principle of locality. It's because you see those equations only involve the value of electric fields and the magnetic field at a single point. If $\vec{B}$ is at $\vec{x}$ point, then $\vec{E}$ is also at $\vec{x}$ point. You never say $\vec{B}_{\vec{x}}$ but $\vec{E}_{\vec{y}}$. So this reflects the principle of locality that the effect is propagating from point to point, you just have the derivative of the same point. You don't involve separate points. So we call Maxwell's equations as local equations, they contain only E and B at same point with finite number of derivatives. So the derivatives are the ones to help you to propagate because it relates the point to the neighboring point.

Another example of locality is the Einstein's general relativity. In Einstein gravity, the dynamical variables are so-called spacetime metric $g_{\mu\nu}(\vec{x}, t)$. They are object with two indices, two spacetime indices, and the Einstein's equations are equations for this kind of object $g_{\mu\nu}$. And again this equation are local equations, in the sense that they only depends on the g evaluated at the same point with a finite number of derivatives. So in Einstein's gravity, you no longer have action at distance, so the effect of the gravity is propagated through the spacetime.

Here we say these two examples examplify the principle of locality. In fact, the principle of locality plays a very important role in formulating those equations, which is a very powerful principle.